CHAPTER 1

A Test-Driven Fable

Imagine two programmers working on the same task. Both are equally skilled, charming, and delightful people, motivated to do a high-quality job as quickly as possible. The task is not trivial, but not wildly complex either. For the sake of discussion, let's say it's a new user logging in to a website and entering some detailed pertinent information.

The first developer, who we'll call Sam, says, "This is pretty easy, and I've done it before. I don't need to write tests." And in five minutes Sam has a working method ready to verify.

Our second developer is named Jamie. Jamie says, "I need to write some tests." Jamie starts writing a test describing the desired behavior. The test is executable and passes if the corresponding code matches the test. Writing the test takes about five minutes. Five additional minutes later, Jamie also has a working method, which passes the test and is ready to verify. Because this is a fable, we're going to assume that Sam is allergic to automated testing, while Jamie is similarly averse to manually verifying against the app in the browser.

At this point you might expect me to say that even though it has taken Jamie more time to write the method, Jamie has written code that is more likely to be correct, robust, and easy to maintain. That's true. I'm going to say that. But I'm also going to say that there's a good chance Jamie will be done before Sam even though Jamie is taking on the additional overhead of writing tests.

Let's watch our programmers as they keep working. Sam has a five-minute lead, but both of them need to verify their work. Sam needs to test in a browser; I said the task requires a user to log in. Let's say it takes Sam one minute to log in and perform the task to verify the code in a development environment. Jamie verifies by running the test, which takes about 10 seconds. (At this point Jamie has to run only one test, not the entire suite.)